

convert json array to csv files in Nifi

Asked 1 month ago Active 1 month ago Viewed 71 times

I have complex json response after InvokeHTTP which needs to be converted to multiple csv files using NiFi. For example below data needs to be converted to three csv files. First csv file will have ID, Owner, address. Second csv will have ID, contact, firstName. Third will have ID, workList, owner.

```
"data" : [ {
  "ID": "789379832",
  "CreatedDate" : "2021-11-13T14:25:17-05:00",
  "Owner" : "abc",
  "workList" : "Sing, Play",
  "address" : "",
  "Phone" : "",
  "contact" : [ "Phone", "Email" ],
  "email" : "",
  "firstName" : "Kim",
  "notes" : "Hello abc",
  "yearCreated" : 2021
}, {
  "ID": "999079832",
  "CreatedDate" : "2021-04-13T09:36:51-04:00",
  "Owner" : "xyz",
  "workList" : "Sing",
  "address" : "80 Hastings Street",
  "Phone" : "",
  "contact" : [ "Phone", "Self" ],
  "email" : "",
  "firstName" : "Hari",
  "notes" : "Hello Hari",
  "yearCreated" : 2021
}
]
```

Example csv1

```
ID,Owner,address
789379832,abc,
999079832,xyz,80 Hastings Street
```

I am using "EvaluateJsonPath" to get key "data" using \$.data. Then to make the json array to be fileflow content using "ReplaceText" and then "SplitJson" array. Then I am using "ConvertRecord" processor with 'JsonTreeReader' and 'CsvRecordSetWriter'. I want to understand how to set attribute values for each csv file in ConvertRecord here. Is my approach not optimal should i use JoltTransformJSON instead.

csv apache-nifi

When you split complex JSON into multiple JSONs and want to use a different schema for each chunk to validate/convert, then you can use Schema Registry as a central repository for all the schema and make validation/convert dynamic. Have a look at this to get an idea -

stackoverflow.com/questions/66367287/... – Vikramsinh Shinde Aug 27 at 7:05

1 Answer

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 Edit:

- 1 Ok so you want to make 3 CSV files, where each Array item is a row in the CSV.
-  Very similar to my original suggestion, but using EvaluateJsonPath instead of ForkRecord.
-  You don't need to be using FlowFile attributes for any of this flow.
- 
 1. EvaluateJsonPath with Destination property set to flowfile-content. This avoids the need for ReplaceText.
 2. 3 ConvertRecords and have the EvaluateJsonPath success relationship into all 3 (each one gets a copy of the records, for your 3 CSVs). Each UpdateRecord has a different CSV Writer with a reduced schema to drop the fields you don't want for that CSV. Then have the UpdateRecords success relationship into PutFile.

Set each CSVRecordSetWriter with a schema with only the fields you want, e.g.

```
{
  "type": "record",
  "name": "CSV",
  "fields" : [
    {"name": "ID", "type": "string"},
    {"name": "Owner", "type": "string"},
    {"name": "address", "type": "string"}
  ]
}
```

So that it ignores all of the fields you don't want.

[Docs for avro schemas](#)

[Tool for validating avro schemas](#)

Example Avro Schema with an Array (for the contacts field)

```
{
  "type": "record",
  "name": "CSV",
  "fields" : [
    {"name": "ID", "type": "string"},
    {"name": "contact", "type":{"type": "array", "items" : "string" }},
    {"name": "firstName", "type": "string"}
  ]
}
```

Original:

So firstly, when you say this For example below data needs to be converted to three csv files - do you mean that each Object in the Array needs to create 3 CSV files? I.e. the example data would produce 3 CSVs for Array item 1, and 3 CSVs for Array item 2?

In this case, I think [ForkRecord](#) is a better fit for you, as you can achieve most of what you're doing in a single processor - you can process the elements of the array as individual records (see the [Additional Details](#) page for examples). With a JSON Reader and a CSV Writer you've already handled the conversion and splits.

After that, you can put down 3 ConvertRecords and have the ForkRecord success relationship into all 3 (each one gets a copy of the records, for your 3 CSVs). Each UpdateRecord has a different CSV Writer with a reduced schema to drop the fields you don't want for that CSV. Then have the UpdateRecords success relationship into PutFile.

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edited Aug 27 at 19:32

answered Aug 27 at 8:47



Sdairs

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entire json array will produce just 3 csv. Edited the question – [likeGreen](#) Aug 27 at 11:35

Updated my answer – [Sdairs](#) Aug 27 at 18:04

thanks this is what I did. But, I didn't use EvaluateJsonpath, I used Funnel to forward to 3 ConvertRecords. Onething, what would be schema for "contact" : ["Phone","Email"],. I was trying with this { "name" : "contact" , "type" : ["null","string"]}, but getting did not have valid JSON content.

– [likeGreen](#) Aug 27 at 18:18

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- 1 Updated my answer with an example avro schema for an Array type - you can find the docs on it here avro.apache.org/docs/current/spec.html#Arrays – [Sdairs](#) Aug 27 at 19:33

We will be getting [Phone, Self], can't we remove open and close parenthesis. Will this approach not work if we want to remove parenthesis as well. I think then some groovy script needs to be written

– [likeGreen](#) Aug 27 at 20:23
